

Finite-Blocklength Latency Minimization for Uplink and Downlink Communications:

Topic-Split Summary Tables

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1 Naming

Table 1: Document naming

Document label	Meaning
Training Only Method	Direct online optimization during scheduling.
Training + Testing Method	Separate training stage followed by testing-stage inference.
Payload completion	Each user continues until its remaining payload is exhausted.

2 Coverage

2.1 Uplink

Table 2: Uplink coverage used in this document

Scenario	Training Only	Training + Testing	Config
Payload completion	Seed 3	Train [1, 2], test 3	UL payload

2.2 Downlink

Table 3: Downlink coverage used in this document

Scenario	Training Only	Training + Testing	Config
Payload completion	Seed 3	Train [1, 2], test 3	DL payload

3 Latency, Synchronization, and Service

3.1 Uplink

Table 4: Uplink latency, synchronization, and service summary

Metric	Payload completion	
	Train Only	Train+ Test
Init. total	0.030733	0.030733
Final total	0.024667	0.025200
Red. (%)	19.7397	18.0043

Table 4: Uplink latency, synchronization, and service summary (continued)

Metric	Payload completion	
	Train Only	Train+ Test
Init. avg	0.010244	0.010244
Final avg	0.008222	0.008400
Init. async.	0.016667	0.016667
Final async.	0.015333	0.015333
Async. red. (%)	8.0000	8.0000
Served bits	6000	6000
Skipped blocks	0	0

3.2 Downlink

Table 5: Downlink latency, synchronization, and service summary

Metric	Payload completion	
	Train Only	Train+ Test
Init. total	0.305000	0.305000
Final total	0.035800	0.073533
Red. (%)	88.2623	75.8907
Init. avg	0.101667	0.101667
Final avg	0.011933	0.024511
Init. async.	0.131600	0.131600
Final async.	0.007867	0.046400
Async. red. (%)	94.0223	64.7416
Served bits	6000	6000
Skipped blocks	3	10

4 Radio Quality

4.1 Uplink

Table 6: Uplink radio-quality summary

Metric	Payload completion	
	Train Only	Train+ Test
Init. SNR (dB)	8.1078	8.1078
Final SNR (dB)	8.2627	8.3355
Init. SINR (dB)	-3.3939	-3.3939
Final SINR (dB)	-3.2868	-3.4146

4.2 Downlink

Table 7: Downlink radio-quality summary

Metric	Payload completion	
	Train Only	Train+ Test
Init. SNR (dB)	8.0000	8.0000
Final SNR (dB)	13.9912	9.9357

5 Training and Runtime

5.1 Uplink

Table 8: Uplink training and runtime summary

Metric	Payload completion	
	Train Only	Train+ Test
Train episodes	–	2
Epoch plan	–	max 100 kkt_converged, kkt_converged, kkt_converged
Solve status	–	87 / 92
Feasible queries	–	15.371869
Best mean rate	–	
Train time (s)	95.283	282.902
Test time (s)	0.000	52.887
Total time (s)	95.283	335.790
FB evals	835	7268
Fwd evals	97	284
Opt. steps	–	334
Opt. updates	738	–

5.2 Downlink

Table 9: Downlink training and runtime summary

Metric	Payload completion	
	Train Only	Train+ Test
Train episodes	–	2
Epoch plan	–	120 / 96
Solve status	–	kkt_converged
Feasible queries	–	26 / 51 (0.509804)
Best mean rate	–	17.961083
Best sum rate	–	22.545255
Train time (s)	419.069	311.368
Test time (s)	0.000	1.764
Total time (s)	419.069	313.132
FB evals	3525	0
Fwd evals	1238	118
Opt. steps / updates	2287	193